



WORKSHOP ON MATHEMATICAL MODELING AND SCIENTIFIC COMPUTING – 2026

Dedicated to the Memory of Nikolai Botkin

TECHNICAL UNIVERSITY OF MUNICH, School of Computation,

Information and Technology, Munich, Germany, September 22–25, 2026

WORKSHOP PROGRAM

Time	Tuesday, September 22	Time	Wednesday, September 23	Time	Thursday, September 24	Time	Friday, September 25
		10:00-11:40	Session 3 Chair: Pirmin Schlicke (University of Salzburg, Austria)	10:00-11:40	Session 7 Chair: Christina Kuttler (Technical University of Munich, Germany)	10:00-11:40	Session 10 Chair: Johannes Zimmer (Technical University of Munich, Germany)
		10:00-10:25	Johannes Zimmer , Mario Ayala (Technical University of Munich, Germany) <i>Law of Large Numbers and Path Large Deviations for Environment-Mediated Cooperation</i>	10:00-10:25	Joseph Páez Chávez (Escuela Superior Politécnica del Litoral, Ecuador), Aytül Gökçe (Ordu University, Turkey), Thomas Götz (University of Koblenz, Germany), Burcu Gürbüz (Johannes Gutenberg-University Mainz, Germany) <i>Continuation-Based Analysis of an SIRS Model with Waning Immunity and Revaccination</i>	10:00-10:25	Niklas Boers (Technical University of Munich, Germany) <i>Anticipating Critical Transitions - from Theory to Data-Driven Pragmatism</i>
		10:25-10:50	Anna Maslovskaya , Ivan Shevkun (Innopolis University, Russia), Christina Kuttler (Technical University of Munich, Munich, Germany) <i>Computational Framework for Simulating Quorum-Regulated Growth of Mixed Bacterial Cultures</i>	10:25-10:50	Petr Andriushchenko (Heidelberg University, Heidelberg, Germany), Andrey Kovtanyuk (Technical University of Munich, Germany) <i>Machine Learning Approach to Modeling Blood Flow in Microvessels</i>	10:25-10:50	Iuliia Kurnaeva , Anna Maslovskaya (Innopolis University, Russia) <i>Distributional and Information-Theoretic Decomposition Technique for Characterization of 3D Cell-Protein Interactions</i>
		10:50-11:15	Christina Kuttler (Technical University of Munich, Germany) <i>(How) Do Bacteria Remember? Nonlinear Dynamics in Quorum Sensing Networks</i>	10:50-11:15	Nora Granzow, Moritz Gubitzer, Shiyuan Niu, Tengxiao Gao, Christina Kuttler (Technical University of Munich, Germany), Joachim Draeger (IABG mbH, Germany) <i>Effective Pandemic Management by Using Data-Driven Threshold Binning Methods</i>	10:50-11:15	Md. Faisal Rahman Dhrubo , Hasibul Hasan Shawon , Lamia Jamal , Mim Mony , Mohammad Nyme Uddin (Imam Mohammad Ibn Saud Islamic University, Bangladesh) <i>Predictive Indoor Air Quality Management in Rural Hospitals: A Machine Learning Framework Based on TVOC Forecasting and Adaptive Ventilation Control</i>
		11:15-11:40	Samvel Sarukhanian , Anna Maslovskaya (Innopolis University, Russia), Christina Kuttler (Technical University of Munich, Germany) <i>Computational Approach for Exploring the Characteristics of Quorum-Mediated Signaling Dynamics during Bacterial Biofilm Growth</i>	11:15-11:40	Providence Nahimana (University of Delhi, India) <i>Energy Consumption Forecasting and Cost Optimization in Smart Buildings Using Simulation-Based Numerical Analysis</i>	11:15-11:40	Ngo Thuan Du (Ho Chi Minh City University of Economics and Finance, Viet Nam) <i>Negative Binomial Model for Clinical Abnormality Index: Comparing Frequentist, Bootstrap-Based Predictive Assessment and Hierarchical Bayesian Approaches</i>
		11:40-13:00	LUNCH BREAK	11:40-13:00	LUNCH BREAK	11:40-12:30	YOUNG SCIENTIST AWARDS & CLOSING CEREMONY
		13:00-14:40	Session 4 Chair: Anna Maslovskaya (Innopolis University, Russia)	13:00-14:15	Session 8 Chair: Petr Andriushchenko (Heidelberg University, Germany)		
12:00-14:00	REGISTRATION	13:00-13:25	Veronika Hofmann (Technical University of Munich, Germany), Pirmin Schlicke (University of Salzburg, Austria), Jan Zawallich (Clausthal University of Technology, Germany), Heiko Enderling (The University of Texas MD Anderson Cancer Center, USA), Christina Kuttler (Technical University of Munich, Germany) <i>Spectral Spatial Analysis of Cancer Biopsies: Validation with In-Silico Data</i>	13:00-13:25	Denis Belomestny , Ekaterina Morozova (Duisburg-Essen University, Germany) <i>Statistical Inference for Degenerated Particle Systems and McKean Vlasov Stochastic Differential Equations with Deep Neural Networks</i>		
		13:25-13:50	Andrey Kovtanyuk (Technical University of Munich, Germany), Aleksandra Gomziakova , Valentin Andreev (Far Eastern Federal University, Russia) <i>2D Agent-Based Modeling of Tumor Cell Evolution</i>	13:25-13:50	Vitalii Kapitan (National University of Singapore, Singapore), Danil Parkhomenko , Pavel Ovchinnikov , Konstantin Soldatov (Far Eastern Federal University, Russia), Gennady Chitov (Universite de Sherbrooke, Canada) <i>Uncovering Hidden Order in Directed Percolation with Deep Neural Networks</i>		
14:00-14:10	OPENING REMARKS Chair of the Program Committee Johannes Zimmer (Technical University of Munich, Germany)	13:50-14:15	Pirmin Schlicke (University of Salzburg, Austria) <i>Mathematical Modeling in Clinical Care</i>	13:50-14:15	G. M. Rahad, Kamroon Nahar, Anasua Pramanik, Tashnim Jahan, Jobayer Hossain (International University of Business Agriculture and Technology, Bangladesh) <i>Deep Learning-Based Solution for Thermal Comfort Prediction in Lobby Spaces of a Multistoried Educational Building</i>		
14:10-15:25	Session 1 Chair: Varvara Turova (Technical University of Munich, Germany)	14:15-14:40	Grace D'Agostino (University of Guelph, Canada) <i>Towards a Mechanistic Model of the Mercury Methylation Process</i>	14:15-14:40	Shubhangi Dwivedi (VIT Vellore, India) <i>Fear of Native Prey Results Synchrony and Brings Balance To a Chaotic Food Chain in Presence of Predation-Driven Apparent Competition</i>		
14:10-14:35	Yuri Ledyayev (Western Michigan University, USA) <i>Lagrange Problem and Optimal Control</i>	14:40-15:00	COFEE BREAK	14:40-15:00	COFEE BREAK		
14:35-15:00	Vladimir Turetsky (Braude College of Engineering, Israel), Valery Glizer (Braude College of Engineering, Israel) <i>Towards Computing Suboptimal Controls in One Class of Infinite-Horizon Cheap Control Linear-Quadratic Differential Games: Asymptotic Analysis Approach</i>	15:00-16:15	Session 5 Chair: Andrey Kovtanyuk (Technical University of Munich, Germany)	15:00-16:15	Session 9 Chair: Irina Sidorenko (Technical University of Munich, Germany)		
15:00-15:25	Alexey Golubev (Ishlinsky Institute for Problems in Mechanics RAS, Russia) <i>Estimation of State Variable Absolute Values Maxima for Feedback Linearizable Dynamical Systems</i>	15:00-15:25	Mario Ayala , Johannes Zimmer (Technical University of Munich, Germany) <i>Derivation of the Symmetric Exclusion Process from a Monitored Bose-Hubbard Model</i>	15:00-15:25	Petr Andriushchenko , Till Bärnighausen (Heidelberg University, Germany) <i>Clustering Methodology for Mixed-Variable Health Surveys</i>		
15:25-15:40	COFEE BREAK	15:25-15:50	Brigitte Forster-Heinlein (University of Passau, Germany) <i>Complexification - Gabor's Analytic Signal Revisited</i>	15:25-15:50	Md Al Hossain Mukib , Sultan Salauddin , Shahed Bin Hossan Riyan , Emon Kumar Nath , Refat Hasan Milon , Md. Hasan (Building Energy & Environmental Management- BEEM Enhanced by AI, Bangladesh) <i>Machine Learning-Based Prediction of Indoor Relative Humidity in University Laboratories Using Regression Models</i>		
15:40-16:55	Session 2 Chair: Alexey Golubev (Ishlinsky Institute for Problems in Mechanics RAS, Russia)	15:50-16:15	Pavel Aronov , Mikhail Galanin , Aleksandr Rodin (Keldysh Institute of Applied Mathematics RAS, Russia) <i>A Combined Domain Decomposition and Lagrange Multiplier Method for Multi-Contact Inelastic mechanics</i>	Thermo- 15:50-16:15	Firuja Tasneem , Rifah Tamanna , Mohammad Nyme Uddin , Ronyma Bormon , Samia Rahman Anamika , Odrita Haque Odhi and Sayma Akter (Building Energy & Environmental Management- BEEM Enhanced by AI, Bangladesh) <i>A BIM-Integrated Explainable Machine Learning Framework for CO₂ Prediction in Naturally Ventilated University Cafeterias</i>		
15:40-16:05	Alessandro Rossi (Sapienza University of Rome, Italy), Patrick Piprek (Airbus Defence and Space GmbH, Germany), Antonio Genova (Sapienza University of Rome, Italy) <i>Model-Predictive-Control-Based Trajectory Generation Enhanced with Reinforcement Learning for Formation Maneuvering</i>	16:15-16:30	COFEE BREAK	16:15-21:00	SOCIAL PROGRAM		
16:05-16:30	Anton Mikhailov (Krasovskii Institute of Mathematics and Mechanics UrB RAS, Russia) <i>Trajectory Simulation for Conflict-Controlled Systems with Multidimensional Control Sets</i>	16:30-17:45	Session 6 Chair: Johannes Diepolder (Diepolder Dynamic Optimization GmbH, Germany)				
16:30-16:55	Varvara Turova (Technical University of Munich, Germany) <i>Some Biomedical Applications of Differential Games</i>	16:30-16:55	Patrick Piprek , David Schwalb (Airbus Defence and Space GmbH, Germany) <i>Preference Incorporation into Generalized Multi-Objective Island Model for Flight Control System Clearances</i>				
		16:55-17:20	Arseniy Spiridonov (Krasovskii Institute of Mathematics and Mechanics UrB RAS, Russia) <i>Conflict-Free Aircraft Flows Merging in Real-Time Airspace Situation</i>				
		17:20-17:45	Viktor Sinitsyn , Florian Holzapfel (Technical University of Munich, Germany) <i>A Protocol-Agnostic, Human- and Machine-Readable ICD Format for Automated Software Generation, Validation, and Verification</i>				